Junyi Zhu

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About Me

I am currently a Senior Researcher at Samsung Research UK (SRUK). I received my PhD from the ESAT Department at KU Leuven in Belgium, where I specialized in artificial intelligence under the supervision of Prof. Matthew Blaschko. Prior to that, I earned my M.S. degree from the Karlsruhe Institute of Technology in Germany, where I worked on several deep learning projects related to autonomous driving at the Institute of Measurement and Control Systems (MRT).

Conducting research in AI and applying these technologies to advance industry and everyday life are my greatest passions. I am actively engaged in exploring recent breakthroughs in the field, both through my own research and by closely following cutting-edge developments. I look forward to the day when AGI becomes a reality.

Employment

Since Sept. 2024 — **Senior Researcher.** Advanced AI research group at Samsung Research UK.

- Led the research and prototyping of personalized on-device AI systems, mobile language models, and distributed training systems for mobile platforms.

Apr. 2024 – Sept. 2024 — **Research Associate.** ESAT-PSI lab of KU Leuven

- Led research on AI search systems and the efficiency of training and inference in image generative models.

Industry Experience

Oct. 2023 - Now	External collaboration with Microsoft Research Lab, Redmond, USA.
Feb. 2024 - Now	External collaboration with MemTensor, Shanghai, China.
Aug. 2023 - Now	External collaboration with Infinigence-AI, Beijing, China.
Nov. 2017 - Feb. 2018	Internship at Hella KGaA Hueck & Co, Lippstadt, Germany.

Academic Engagements

2025	Area chair at CVPR.
Aug. 2023 - Oct. 2023	Visiting PhD student in the NICS-EFC lab at Tsinghua University.
June 2023	Attended Generative Modeling Summer School (GeMSS).
July 2022	Attended Machine Learning Summer School (MLSS).
2021 - 2025	Reviewer at ICML, NeurIPS, ICLR, CVPR, ICCV, ECCV, TMLR, AISTATS, etc.
Aug. 2019 - Mar. 2020	Research assistant in MRT institute in Karlsruhe.
Apr. 2018 - Apr. 2019	Student researcher in FZI research center in Karlsruhe.

Talks

May 2024 — **Venue:** ICLR Tiny Paper Track
Topic: Rescaling Intermediate Features Makes Trained Consistency Models Perform Better.

Nov. 2023 — **Venue:** Leuven.AI Junior Researchers Day.
Topic: Confidence-aware Personalized Federated Learning via Variational Expectation Maximization.

Dec. 2022 — **Venue:** NeurIPS WorkShop.
Topic: Tackling Personalized Federated Learning with Label Concept Drift via Hierarchical Bayesian Modeling.

Research Publications

1 *Wang, Z., ***Zhu, J.**, *Tang, B., Li, Z., Xiong, F., Yu, J., Blaschko, M.B. (2025). Jigsaw-R1: A Study of Rule-based Visual Reinforcement Learning with Jigsaw Puzzles. arXiv.

2 Lin, Z., Liu, E., Ning, X.,**Zhu. J**, Wang W., Yekhanin, S. (2025). Latent Zoning Network: A Unified Principle for Generative Modeling, Representation Learning, and Classification. The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS).

3 **Zhu, J.**, Ozkan, S., Maracani, A., Mutlu S., Min C., Ozay, M. (2025). Multi-Task Pre-Finetuning of Lightweight Transformer Encoders for Text Classification and NER. The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP).

4 **Zhu, J**, Yao, R., Ceritli, T., Ozkan, S., Blaschko, K. B., Noh, E., Min, J., Min, C.J., Ozay, M. (2025). Guided Model Merging for Hybrid Data Learning: Leveraging Centralized Data to Refine Decentralized Models. The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV).

5 Wang, Y., Wang, P., Xi, C, Tang, B., **Zhu, J**, Wei, W., Chen, C., Yang, C., Zhang, J., Lu, C., Niu, Y., Mao, K., Li, Z., Xiong, F., Hu, J., Yang, M. (2025). Adversarial Preference Learning for Robust LLM Alignment. Findings of Annual Meeting of the Association for Computational Linguistics (ACL).

6 *Tang, B., ***Zhu, J.**, Wang, H., Xi, C., Ge, Y., Wu, J., Ge, T., Jin, B., Feng, Y., Niu, Y., Wei, W., Yu, Y. Li, C., Lin, Z., Wu, H., Liao, N., Yang, Y., Wang, J., Li, Z., Xiong, F., Chen, J. (2025). Xinyu AI Search: Enhanced Relevance and Comprehensive Results with Rich Answer Presentations. arXiv.

7 *Liu, E., ***Zhu, J.**, Lin, Z., Ning, X., Blaschko, M. B., Yekhanin, S., Yan, S., Dai, G., Yang, H., Wang, Y. (2025). Linear Combination of Saved Checkpoints Makes Consistency and Diffusion Models Better. In International Conference on Learning Representations (ICLR).

8 ***Zhu, J.**, *Liu, S., Yu, Y., Tang, B., Yan, Y., Li, Z, Xiong, F., Blaschko, M. B. (2024). FastMem: Fast Memorization of Prompt Improves Context Awareness of Large Language Models. Findings of Conference on Empirical Methods in Natural Language Processing (EMNLP).

9 *Liu, E., ***Zhu, J.**, Lin, Z., Ning, X., Blaschko, M. B., Yan, S., Dai, G., Yang, H., Wang, Y. (2024). Efficient Expert Pruning for Sparse Mixture-of-Experts Language Models: Enhancing Performance and Reducing Inference Costs. arXiv.

10 *Shi, J., ***Zhu, J.**, Pelt, D, Joost, B, Blaschko, M. B. (2024). Implicit Neural Representations for Robust Joint Sparse-View CT Reconstruction. Transactions on Machine Learning Research (TMLR).

11 **Zhu, J.**, Lin, Z., Liu, E., Ning, X., Blaschko, M. B. (2024). Rescaling intermediate features makes trained consistency models perform better. In the second Tiny Papers Track at ICLR (notable).

12 **Zhu, J.**, Yao, R., Blaschko, M. B. (2023). Surrogate model extension (SME): A fast and accurate weight update attack on federated learning. In Proceedings of the 40th International Conference on Machine Learning (ICML).

13 **Zhu, J.**, Blaschko, M. B. (2023). Improving differentially private SGD via randomly sparsified gradients. Transactions on Machine Learning Research (TMLR).

14 ***Zhu, J.**, *Ma, X., Blaschko, M. B. (2023). Confidence-aware personalized federated learning via variational expectation maximization. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

15 *Ma, X., ***Zhu, J.**, Blaschko, M. B. (2022). Tackling personalized federated learning with label concept drift via hierarchical Bayesian modeling. In Workshop on Federated Learning: Recent Advances and New Challenges (in conjunction with NeurIPS) (oral).

16 **Zhu, J.**, Blaschko, M. B. (2021). R-GAP: Recursive gradient attack on privacy. In International Conference on Learning representations (ICLR).

17 Hu, H., **Zhu, J.**, Wirges, S., Lauer, M. (2019). Localization in aerial imagery with grid maps using locgan. In IEEE Intelligent Transportation Systems Conference (ITSC).

18 Kamran, D., **Zhu, J.**, Lauer, M. (2019). Learning path tracking for real car-like mobile robots from simulation. In European Conference on Mobile Robots (ECMR) (oral).

[* indicates co-first authorship.]

Education

Ph.D. **KU Leuven, Leuven, Belgium**
Advised by Prof. Matthew Blaschko at ESAT-PSI lab.

Dissertation: Privacy-Preserving and Distributed Machine Learning.

PhD Scholarship (four-year): funded by the Flemish Government (AI Research Program) and the Research Foundation – Flanders (FWO).

Additional research topics during PhD include: 1) Instruction-following and inference efficiency of language models. 2) Training and inference efficiency of diffusion models. 3) Image and scene reconstruction using implicit neural representation.

Education (continued)

M.Sc.	**Karlsruhe Institute of Technology, Karlsruhe, Germany** ***Thesis***: Transferring Grid Maps to Aerial Imagery for Localization in Autonomous Driving System using Generative Adversarial Network.
B.Sc. (Exchange Program)	**Dresden University of Technology, Dresden, Germany** ***Thesis***: Design and Implementation of Automated Penetration Measuring System.
B.Sc.,	**Beijing Institute of Technology, Beijing, China**